

SOLIDWORKS COURSE CONTENT

Duration- 6 Weeks

WEEK - 01

Introduction to SolidWorks & Basic Sketching

- Overview of CAD and SolidWorks Interface
- Understanding design intent
- Creating 2D sketches
- Sketch relations and dimensions
- Sketch tools: Line, Rectangle, Circle, Arc, Trim, Mirror
- Practice: Simple 2D sketching exercises

WEEK - 02

Part Modeling Basics

- Introduction to 3D modeling
- Feature-based modeling (Extrude, Revolve, Cut)
- Working with planes and reference geometry
- Editing and modifying features
- Practice: Create basic 3D parts (e.g., bracket, pulley)

WEEK - 03

Advanced Part Modeling

- Fillet, Chamfer, Shell, Rib, Draft
- Patterns: Linear, Circular, Mirror
- Configurations and design table
- Introduction to equations in design
- Practice: Model complex parts (e.g., connecting rod, gear)

WEEK - 04

Assemblies

- Introduction to assemblies
- Mates: Standard, Advanced, Mechanical
- Subassemblies and exploded views
- Interference detection and collision analysis
- Practice: Assemble a mechanical system (e.g., piston-cylinder)

WEEK - 05

Drawing and Detailing

- Creating 2D drawings from parts and assemblies
- Views: Standard, Section, Detail, Auxiliary
- Dimensioning and tolerancing
- BOM (Bill of Materials) and annotations
- Practice: Generate production drawings

WEEK - 06

Applications & Project Work

- Introduction to simulation (basic stress analysis)
- Motion study basics
- Design validation and optimization tools
- Project assignment: Design and model a mechanical product (e.g., gearbox, robotic arm)
- Final assessment and feedback